

Learning4help by Waqas Qureshi

Asslam-o-Alaikum! In this file we'll provide you **PHY101 Important Midterm MCQs. Must Prepare before Midterm Exams.**
Verify Answers yourself too. I am not responsible for any answer. If you found mistake then inform me.
If you found mistake then let us know.

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Question # 1 of 10 (Start time: 08:33:43 PM, 07 November 2024) Total Marks: 1

The product of force and time is called change in:

Select the correct option

- ☐ Force
- ☒ Impulse
- ☐ Torque
- ☐ Acceleration

PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 2 of 10 (Start time: 08:34:04 PM, 07 November 2024) Total Marks: 1

A mass of 5 kg is lifted to a height of 10 meters. What is the gravitational potential energy gained by the mass?

Select the correct option

- ☐ 100 J
- ☐ 50 J
- ☐ 200 J
- ☒ 500 J

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 3 of 10 (Start time: 08:34:30 PM, 07 November 2024) Total Marks: 1

In the absence of an applied force (F) on an object, what conclusion can be drawn about the object's power?

Select the correct option

☐ Negative

☐ Positive

☐ Indeterminate

☒ Zero

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 4 of 10 (Start time: 08:35:19 PM, 07 November 2024) Total Marks: 1

What happens to kinetic energy in a typical inelastic collision?

Select the correct option

☒ It is converted to other forms of energy.

☐ It is conserved.

☐ It doubles.

☐ Can not be determined.

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 5 of 10 (Start time: 08:35:45 PM, 07 November 2024) Total Marks: 1

A car with a mass of 1500 kg accelerates at a rate of 6 m/s^2 . Calculate the force required to produce this acceleration.

Select the correct option

☐ 0.090 N

☒ 9000 N

☐ 0.900 N

☐ 900 N

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 6 of 10 (Start time: 08:36:12 PM, 07 November 2024) Total Marks: 1

As an object moves away from the Earth's surface, what happens to the value of 'g' (acceleration due to gravity)?

Select the correct option

☐ It remains constant

☐ It increases

☒ It decreases

☐ It becomes zero

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Question # 7 of 10 (Start time: 08:36:40 PM, 07 November 2024) Total Marks: 1

Newton's second law of motion is also called:

Select the correct option

- ☐ Law of gravitation
- ☒ Law of acceleration
- ☐ Law of inertia
- ☐ Law of torque

Question # 8 of 10 (Start time: 08:37:08 PM, 07 November 2024) Total Marks: 1

A bullet shoot straight up returns to its starting point in 10 sec, the initial speed was:

Select the correct option

- ☐ 98 m/s
- ☐ 9.8 m/s
- ☒ 49 m/s
- ☐ 24.5 m/s

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 9 of 10 (Start time: 08:37:55 PM, 07 November 2024) Total Marks: 1

In order to find the density of an object , which two measurements are required?

Select the correct option

☐ mass and length

☐ temperature and volume

☐ volume and length

☒ mass and volume

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PHY101 - Physics (1st graded quiz PHY101) Quiz Start Time: 08:33 PM, 07 November 2024

Question # 10 of 10 (Start time: 08:39:03 PM, 07 November 2024) Total Marks: 1

How is the scalar product of two vectors A and B calculated?

Select the correct option

☐ $A \times B = |A| \times |B| \times \sin(\theta)$, where θ is the angle between A and B

☐ $A \cdot B = |A| \times |B| \times \sin(\theta)$, where θ is the angle between A and B

☒ $A \cdot B = |A| \times |B| \times \cos(\theta)$, where θ is the angle between A and B

☐ $A \cdot B = |A| \times |B| \times \tan(\theta)$, where θ is the angle between A and B

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Question # 1 of 10 (Start time: 10:07:51 PM, 16 November 2024) Total Marks: 1

In _____ object moves farther away from its original position if displaced slightly

Select the correct option

- ☒ unstable equilibrium
- ☐ stable equilibrium
- ☐ rotational equilibrium
- ☐ dynamic equilibrium

PHY101 – Physics (2nd graded quiz PHY101) Quiz Start Time: 10:07 PM, 16 November 2024

Question # 2 of 10 (Start time: 10:08:07 PM, 16 November 2024) Total Marks: 1

Two forces, 3 N and 4 N, act on an object at right angles to each other. What is the magnitude of the resultant force?

Select the correct option

- ☐ 12 N
- ☒ 5 N
- ☐ 1.33 N
- ☐ 7 N

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PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:07 PM, 16 November 2024

Question # 3 of 10 (Start time: 10:08:44 PM, 16 November 2024) Total Marks: 1

What does the derivative d^2x/dt^2 represent in terms of motion?

Select the correct option

☐ Velocity

☒ Acceleration

☐ Position

☐ Speed

Reload Math Equations

Question # 4 of 10 (Start time: 10:09:18 PM, 16 November 2024) Total Marks: 1

A particle oscillating in simple harmonic motion is:

Select the correct option

☐ never in equilibrium because there is always a force

☐ in equilibrium at the ends of its path because its velocity is zero there

☐ in equilibrium at the ends of its path because its velocity is zero there

☒ in equilibrium at the center of its path because the acceleration is zero there

☐ never in equilibrium because it is in motion

Question # 5 of 10 (Start time: 10:09:49 PM, 16 November 2024) Total Marks: 1

Which of the following statement/s is / are true?

- i. The weight of a man on the Moon is smaller than that on the Earth.
- ii. The mass of a man is the same on both the Moon and the Earth.
- iii. We cannot determine our own mass in the outer space because there is no gravity.

Select the correct option

- ☐ (i) only
- ☐ (ii) and (iii) only
- ☒ (i) and (ii) only
- ☐ (iii) only

Question # 6 of 10 (Start time: 10:10:25 PM, 16 November 2024) Total Marks: 1

A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After collision, the velocity of light particle will be:

Select the correct option

- ☒ 10 m/s
- ☐ 5 m/s
- ☐ 20 m/s
- ☐ 2.5 m/s

PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:07 PM, 16 November 2024

Question # 7 of 10 (Start time: 10:11:10 PM, 16 November 2024) Total Marks: 1

How are positive angles conventionally measured in the cartesian plane?

Select the correct option

Reload Math Equations

- ☒ Counterclockwise from the positive x-axis.
- ☐ Clockwise from the positive y-axis.
- ☐ Counterclockwise from the positive y-axis.
- ☐ Clockwise from the positive x-axis.

Question # 8 of 10 (Start time: 10:11:50 PM, 16 November 2024) Total Marks: 1

The center of mass of the system consisting of Earth, the Sun, and the planet Mars is:

Select the correct option

- ☒ closer to the Sun than to either of the other bodies.
- ☐ at the geometric center of the triangle formed by the three bodies.
- ☐ closer to Mars than to either of the other bodies.
- ☐ closer to Earth than to either of the other bodies.

Question # 9 of 10 (Start time: 10:12:39 PM, 16 November 2024) Total Marks: |

A body is changing its direction but its speed is constant, the object is:

Select the correct option

- ☒ accelerating
- ☐ speeding
- ☐ Displacing
- ☐ slowing down

PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:07 PM, 16 November 2024

Question # 10 of 10 (Start time: 10:13:04 PM, 16 November 2024) Total Marks: |

In a bicycle, the purpose of the chain is to:

Select the correct option

- ☐ Maintain constant force throughout
- ☒ Ensure uniform rotation speed of both gears
- ☐ Change the direction of rotation
- ☐ Increase the wheel's radius

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PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 1 of 10 (Start time: 10:14:44 PM, 16 November 2024) Total Marks: 1

How does the period (T) change if the the linear speed (v) is doubled while keeping radius (R) of the circular path constant?

Select the correct option

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- ☐ Cannot be determined
- ☐ Remains the same
- ☐ Increases
- ☒ Decreases

PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 2 of 10 (Start time: 10:15:27 PM, 16 November 2024) Total Marks: 1

When a particle moving along a circular path, its projection along the diameter executes:

Select the correct option

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- ☐ Vibratory motion
- ☐ Linear motion
- ☒ Simple harmonic motion
- ☐ Rotatory motion

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An inelastic collision is one in which:

Select the correct option

- ☒ momentum is conserved but kinetic energy is not conserved
- ☐ neither kinetic energy nor momentum is conserved
- ☐ momentum is not conserved but kinetic energy is conserved
- ☐ total mass is not conserved but momentum is conserved

PHY101 - Physics (2nd graded quiz PHY101) Time Left: 00:00:00 sec(s) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 4 of 10 (Start time: 10:16:50 PM, 16 November 2024) Total Marks: 1

In static equilibrium, what conditions must be satisfied for a beam resting on supports?

Select the correct option

- ☐ Only demand that the total torque vanishes
- ☒ Both balance forces in the vertical direction and demand that the total torque vanishes
- ☐ Ignore torque considerations
- ☐ Only balance forces in the vertical direction

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PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 5 of 10 (Start time: 10:17:10 PM, 16 November 2024) Total Marks: 1

A lift is accelerating upwards. If the gravitational force is 9.8 m/s^2 , what will a passenger feel in terms of weight?

Select the correct option

☒ Heavier than usual

☐ Lighter than usual

☐ The same as usual

☐ No weight at all

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PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 6 of 10 (Start time: 10:17:40 PM, 16 November 2024) Total Marks: 1

Assess the stability of a rigid body in neutral equilibrium.

Select the correct option

☐ It moves farther away from the original position

☐ It returns to the original position

☐ It is always unstable

☒ It stays in its new position

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Force:

Select the correct option

- ☐ is the ability to do work
- ☐ equals the negative integral (with respect to distance) of the potential energy function
- ☒ equals the time rate of change of momentum
- ☐ is the rate of change of doing work

PHY101 - Physics (2nd graded quiz PHY101) sec(s) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 8 of 10 (Start time: 10:18:27 PM, 16 November 2024) Total Marks: 1

A stable position is achieved when the center of gravity is:

Select the correct option

- ☒ Lowest relative to the object's orientation
- ☐ Constantly shifting
- ☐ Below the supporting surface
- ☐ Located outside the object

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Question # 9 of 10 (Start time: 10:18:49 PM, 16 November 2024) Total Marks: 1

Lighter objects always reach the ground later than heavier objects.
Which is the most suitable explanation for the above statement?

- i. There are some external forces acting on the objects, for example, the air friction.
- ii. A heavier object has a larger uniform velocity when it falls freely.
- iii. A heavier object has a larger acceleration due to gravity.

Select the correct option

- ☐ (ii) and (iii) only
- ☐ (i), (ii) and (iii)
- ☐ (ii) only
- ☒ (i) only

PHY101 - Physics (2nd graded quiz PHY101) Quiz Start Time: 10:14 PM, 16 November 2024

Question # 10 of 10 (Start time: 10:19:12 PM, 16 November 2024) Total Marks: 1

What is tension in a string or rope when an object is in equilibrium?

Select the correct option

- ☐ Varies depending on the object's shape
- ☐ Greater than the object's weight
- ☒ Equal to the object's weight
- ☐ less than the object's weight

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 1 of 10 (Start time: 02:14:24 PM, 29 November 2024) Total Marks: 1

Which of the following scenarios is NOT an example of equilibrium

Select the correct option

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☐ A stone at rest on the ground

☒ A car accelerating

☐ A ladder leaning against a wall

☐ An aircraft flying at a constant speed

Click to Save Answer & Move to Next Question

PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 2 of 10 (Start time: 02:14:57 PM, 29 November 2024) Total Marks: 1

What is the range of possible values for the scalar product of two vectors A and B?

Select the correct option

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☐ It is always positive

☐ It is always zero

☐ It is always negative

☒ It can be any real number

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 3 of 10 (Start time: 02:15:22 PM, 29 November 2024) Total Marks: 1

In the Doppler effect equation for sound, what remains constant while frequency changes?

Select the correct option

☐ Intensity

☐ Wavelength

☐ Amplitude

☒ Speed of sound in the medium

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Click to Show Answer & Move to Next Question

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 4 of 10 (Start time: 02:16:51 PM, 29 November 2024) Total Marks: 1

Why is the decibel scale logarithmic?

Select the correct option

☐ To simplify calculations

☐ To adjust for changes in frequency

☒ To better represent the wide range of human hearing perception

☐ To make it compatible with other logarithmic scales

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 5 of 10 (Start time: 02:17:30 PM, 29 November 2024) Total Marks: 1

If a force acts normally and uniformly from all sides of a body, what type of stress is exerted on the body?

Select the correct option

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☐ Shearing stress

☐ Tensile stress

☐ Tangential stress

☒ Volume stress

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Question # 6 of 10 (Start time: 02:18:13 PM, 29 November 2024) Total Marks: 1

The dimensional units of ratio of work and power is:

Select the correct option

☐ J

☒ T

☐ F

☐ L

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 7 of 10 (Start time: 02:18:51 PM, 29 November 2024) Total Marks: 1

What is the angle at which a body just starts to slide down on an inclined plane called?

Select the correct option

Angle of slip

Angle of inclination

Angle of descent

Angle of repose

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 8 of 10 (Start time: 02:19:10 PM, 29 November 2024) Total Marks: 1

If a rigid body is in static equilibrium, which condition should be satisfied for it?

Select the correct option

Constant torque

Continuous rotational motion

Non-zero linear momentum

Vanishing net torque and net force

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 9 of 10 (Start time: 02:19:51 PM, 29 November 2024) Total Marks: |

An object's speed is measured as 5×10^3 cm/s. Convert this speed to meters per second.

Select the correct option

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☒ 50 m/s

☐ 0.5 m/s

☐ 500 m/s

☐ 5 m/s

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PHY101 - Physics (3rd graded quiz PHY101) Quiz Start Time: 02:14 PM, 29 November 2024

Question # 10 of 10 (Start time: 02:20:53 PM, 29 November 2024) Total Marks: |

A mass of 5 kg is lifted to a height of 10 meters. What is the gravitational potential energy gained by the mass?

Select the correct option

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☒ 500 J

☐ 200 J

☐ 100 J

☐ 50 J

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